



First Semester 2026-27

Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : MEL G643

Course Title : Verification, Test, and Testability of VLSI Circuits

Instructor-in-charge : ABHIJIT ASATI (abhijit_asati@pilani.bits-pilani.ac.in, 2220-Q)

1. **Course description:** VLSI design flow, Simulation based verification, Test bench tools and languages, System Verilog, Formal verification, Impact of scaling, wafer sort or probe test, DC and AC parametric test, Test economics and product quality, Fault models and types; Fault equivalence, Fault dominance, Automated test generation for combinational logic (fault simulation, ATPG, D-algorithm, PODEM, FAN); Test generation for sequential logic; Testability measures, Design for testability (Full-scan and partial scan, BIST, Boundary scan), Delay fault; Analog and mixed signal circuit testing, Memory test, IDDQ test
2. **Scope and Objective:** The objective of this course is to deal with the study of VLSI design flow, Functional verification, verification flow, need of electronic testing, fault modeling, test generation for combinational circuits, test generation for sequential circuits, fault simulation, Built-In Self-Test (BIST), Memory testing, Design for Testability (DFT), SoC test, fault diagnosis, Analog/RF test.
3. **Course learning outcomes:** Students will learn the VLSI design flow, understand the importance of design verification during design flow and also understand the importance of post fabrication testing in VLSI design flow with exposure to the design tools.
4. **Text books and references:**

TB1. B. Wile, John C. Goss and W. Rosner “ Comprehensive Functional Verification” Morgan Kaufmann, 2005.

TB2. Michael. L. Bushnell, and Vishwani. D. Agrawal, “Essentials of Electronic Testing For Digital, Memory and Mixed Signal VLSI Circuits” Kluwer Academic Publishers, Third Edition, 2004

RB1. “ System Verilog for Verification,” Springer Publications, second edition 2008.



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RB2. Stuart Sutherland, Simon Davidmann, Peter Flake” System Verilog for Design,” Springer Publications, Second Edition 2006.

RB3. M Abromovici, M A Breuer & A. D. Friedman "Digital Systems Testing and Testable Design “, Jaico Publications, Paperback Impression, 2001.

RB4. H. Fujiwara, “Logic Testing and Design for testability” MIT Press, 1985.

RB5. Neil H. E. Weste, David Harris and Ayan Banerjee, “CMOS VLSI Design: A circuit and systems perspective.” Pearson Publications, Third Edition 2006.

5. Course Format & Weekly Workflow

Course work involves 3 lecture hour of class work and lab assignment and projects.

6. Course Plan:

Week No.	Module Number	Lecture session	Learning Outcome
1-2	1.	L1.1. Introduction of VLSI test and Testability	Introduction to VLSI test and Testability, low power VLSI design flow. To understand basic VLSI design verification (Power Analysis, Signal Integrity Checks, Physical Verification, Parasitic Extraction, Static Timing Analysis (STA), Formal Verification).
		L 1.2-L1.5 Low power VLSI design flow and verification	
2-5	2	L2.1. Functional verification and verification challenges , Verification Methodology Evolution	To study simulation based functional verification and to understand verification challenges. To learn basic verification environment and techniques. Case study using CALC1 and CALC2 examples.
		L2.2. Verification Environment Basics , Assertions, Testbench and tester, Layered Testbench	
		L2.3. Verification example: CALC 1, Deterministic Verification of Calc1	
		L2.4 CALC2 Overview, Calc2 Verification Plan	



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		L2.5 Verification Coverage	
		L2.6 Strategies for Stimulus Generation	
		L2.7 Deterministic and Random test cases	
		L2.8 Types of Result Checking, Types of self checking test benches	
		L2.9 Coverage results for single Calc2, debug process	
		L2.10 Advanced Verification Methodology	
		L2.11 Re use strategies	
5-7	3	L3.1 High-Level Verification Languages, Introduction to System Verilog	To study different Test bench tools and special purpose languages for test bench writing.
		L3.2 Data Types	
		L3.3-3.4 System Verilog programs	
		L3.5 Basic Object Oriented Programming	
		L3.6 System Verilog programs on OOP	
	4	L4.1- L4.2: Formal verification (FV): BDD, Formal Boolean equivalence checking, property checking, Semi-formal verification	To study formal verification methods. The promise of formal verification (FV) is to achieve stronger, true verification of the DUV instead of merely testing with simulation-based methods.



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9	5	L5.1. Impact of scaling, Test Complexity, Power dissipation, Wafer sort or probe test, DC parametric tests, AC parametric tests	To understand basics of VLSI testing such as fault modeling, Fault equivalence, Fault Dominance and to study classes of stuck-at faults
		L5.2. Test economics and product quality	
		L5.3 Fault modeling, : Stuck-at, bridge, delay, and cross-talk fault, Delay faults	
		L5.4 Fault equivalence, Fault Dominance	
		L 5.5 Classes of Stuck-at Faults Multiple Stuck-at Faults	
9-10	6	L6.1 Event-driven simulation, Compiled-code simulation	To study techniques of fault simulation which determines the coverage of a given set of input stimuli (vectors) for a given fault model or a given fault list. Study of serial fault simulation, parallel fault simulation, deductive fault simulation and concurrent fault simulation.
		L6.2 Serial Fault Simulation, Parallel Fault Simulation	
		L6.3 Deductive Fault Simulation, Concurrent Fault Simulation	
10	7	L7.1 Fault Table Computing, Boolean Difference	Test generation for combinational circuits: ATPG algorithms (Boolean Difference, D-algorithm, PODEM, FAN) and to study testability measures.
		L7.2 D-Algorithm	
		L7.3 Path Oriented Decision Making (PODEM), FAN	
		L7.4 Testability Measures	
11-12	8	L8.1. Internal Scan, Time frame expansion model	Test generation for sequential circuits: Time frame expansion model. Scan chain based DFT technique (Full-scan and partial scan). Design for testability and delay Test.
		L8.2. Cycle-free and cyclic circuits	
		L8.3 Design for Testability (DFT)]	
		L8.4. Full-scan and partial scan methods	
		L8.5 Delay Test	



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13-14	9	L 9.1 Built-In Self-Test (BIST)	To study 'Built in self test' techniques and memory testing methods.
		L 9.2 Memory test, Fault diagnosis	
	10	L10.1 IDDQ Test	To study 'Boundary Scan' Standard and IDDQ testing methods.
		L10.2 Boundary Scan Standard	
	11	L 11.1 Analog and Mixed Signal Test	To study testing methods for analog and mixed signal circuits.

7. Evaluation Scheme:

EC. No.	Evaluation Component	Duration	Weightage	Date & Time	Nature of Component
1.	Mid semester Test	90 Min.	30 %	07/10/2026 FN2	Closed Book
2.	Comprehensive Examination	3 Hrs.	40 %	08/12/2026 AN	Closed Book (20 %) + Open Book (20 %)
3.	Projects	Regular	30 %	TBA	Open Book

* Each student must have minimum 50 % attendance otherwise it will impact their evaluation in EC No. 3.

Criterion for NC: A student should obtain 25 % of the average of the class, or 20 % of the course total marks, whichever is lower to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded NC.

8. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.



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9. **Chamber consultation hours:** TBA

10. **Notices:** Displayed on Nalanda (<https://nalanda-aws.bits-pilani.ac.in>).

11. **Makeup Policy:** Makeup will be given only for genuine cases (accompanied by proof) for Mid semester or comprehensive exam only.

Abhijit Asati

Instructor-in-charge

MEL G626